

Econometrics I
Classical Normal Regression Model
Sampling distributions of the OLS estimators under Normality

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Introduction and Goals

- The ultimate goal is to be able to make statistical inferences about the population parameter vector β
- In particular, we would like to construct statistical hypothesis tests about it.

Is school attendance important?

This example is motivated by the dataset `ATTEND`, which includes a sample about student characteristics from Michigan State University.

$$final = \beta_0 + \beta_1 missed + \beta_2 priGPA + \beta_3 ACT + u$$

where ACT is the achievement test score. We would like to test the hypothesis that missing lectures has no effect on final exam performance, after accounting for prior GPA and ACT^a score.

The null hypothesis, that missing lecture has no effect on final exam performance (after accounting for prior MSU GPA and ACT score), can be written as

$$H_0 : \beta_1 = 0$$

^aTogether with the more well-known SAT, the ACT is another college-entrance standardized test

The Classical Normal Linear Model.

- To be able to construct an hypothesis test, we need to derive the **sampling distribution** of the OLS estimator $\hat{\beta}$.
- There is no easy way to do this unless we introduce an *extra assumption*.
- This extra assumption will be a **distributional assumption about the normality of the errors**.
- In the next slide I recall the previous assumptions MLR1–MLR5, i.e., the Gauss-Markov assumptions.

MLR.1 Linear in parameters

The model can be written as $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}$

MLR.2 Random Sampling

The observations in $\mathbf{y}$ and $\mathbf{X}$ come from a random sample of size n from the population.

MLR.3 No perfect collinearity

The matrix of $\mathbf{X}$ has rank $k + 1$, i.e., there are $k + 1$ linearly independent columns.

MLR.4 Zero Conditional Mean

$$E(\mathbf{u}|\mathbf{X}) = \mathbf{0}.$$

MLR.5 Homoskedasticity

$$V(\mathbf{u}|\mathbf{X}) = \sigma^2 \mathbf{I}_n,$$

To the previous 5, we add the following:

MLR.6 Normality

The population error u is independent of $(x_1, \dots, x_k)$ and is normally distributed with mean zero and variance σ^2 :

$$u \sim N(0, \sigma^2)$$

In matrix form, $\mathbf{u}$ is, multinormal, with mean zero and variance-covariance matrix $\sigma^2 \mathbf{I}$

$$\mathbf{u}|\mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}).$$

MLR6 imposes full independence between $\mathbf{u}$ and $\mathbf{X}$ (not just mean and variance independence), which is where the label of $\mathbf{X}$ as independent variables originated.

The CNR Model

- Assumptions MLR1-MLR6 compose what is called the **Classical Normal Regression Model** or **Classical Linear Model**.
- Notice that the assumptions imply that

$$\mathbf{y}|\mathbf{X} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I}),$$

that is, it implies a joint probability distribution for the random vector $(y, x_1, \dots, x_k)'$ in which the conditional distribution of y given the x 's is

$$y|x_1, \dots, x_k \sim N(\beta_0 + \beta_1 x_1 + \dots + \beta_k x_k, \sigma^2)$$

Sampling distributions.

- We are now in position to derive the practical implications of the CNR model, i.e., to derive the sampling distributions of the OLS statistics $\hat{\beta}$ and $\hat{\mathbf{u}}'\hat{\mathbf{u}}$.
- In here I will derive the sampling distribution of the statistic $\hat{\beta}$ and defer the discussion of the sum of squared residuals $\hat{\mathbf{u}}'\hat{\mathbf{u}}$ until later.
- The key result of the CNR model is that the random vector $\hat{\beta}$ is multnormally distributed.

D1. $\hat{\beta}$ is multinormal

$$\hat{\beta} \sim N(\beta, \sigma^2 \mathbf{Q}^{-1})$$

Proof.

Recall that $\hat{\beta} = \mathbf{A}\mathbf{y}$, where $\mathbf{A} = \mathbf{Q}^{-1}\mathbf{X}'$ is a constant $(k+1) \times n$ matrix. Multiplication by a nonsingular matrix preserves rank, so $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{X}') = \text{rank}(\mathbf{X}) = k+1$. So $\hat{\beta}$ is a full-row-rank linear function of the multinormal vector $\mathbf{y}$ and hence $\hat{\beta}$ is multinormal according to P5 in the slides `MultivariateNormal-distribution.pdf`, slide 9.



Any full-row-rank linear function of the multinormal vector $\hat{\beta}$ will also be multinormal.

D2

Let $\mathbf{t} = \mathbf{H}\hat{\beta}$ and $\boldsymbol{\theta} = \mathbf{H}\beta$, where the $p \times (k+1)$ matrix $\mathbf{H}$ is nonrandom with rank $\text{rank}(\mathbf{H}) = p$. Then $\mathbf{t} \sim N(\boldsymbol{\theta}, \sigma^2 \mathbf{D}^{-1})$, where $\mathbf{D} = (\mathbf{H}\mathbf{Q}^{-1}\mathbf{H}')^{-1}$.

As a special case we have:

D3

Let $\hat{\beta}_j$ be the j th element of $\hat{\beta}$. Then $\hat{\beta}_j \sim N(\beta_j, \sigma^2 q^{jj})$.

Proof.

Take $\mathbf{H} = \mathbf{h}'$ is the $(k+1 \times 1)$ vector with a 1 in the j th position and 0's elsewhere. Then $\mathbf{H}\beta = \beta_j$, $\mathbf{H}\hat{\beta} = \hat{\beta}_j$, $\mathbf{H}\mathbf{Q}^{-1}\mathbf{H}' = q^{jj}$. □

- The result D2 also subsumes other linear functions of the elements of $\hat{\beta}$, such as $\hat{\mu}_i = \mathbf{x}'_i \hat{\beta}$.
- Indeed it subsumes D1, take $\mathbf{H} = \mathbf{I}_k$

Sampling distribution of quadratic forms of $\hat{\beta}$

Now we proceed with quadratic functions of $\hat{\beta}$. By property D2, recall that $\mathbf{t} \sim N(\boldsymbol{\theta}, \sigma^2 \mathbf{D}^{-1})$ where $\mathbf{D} = (\mathbf{H}\mathbf{Q}^{-1}\mathbf{H}')^{-1}$

D4

$$w = (\mathbf{t} - \boldsymbol{\theta})' \mathbf{D} (\mathbf{t} - \boldsymbol{\theta}) / \sigma^2 \sim \chi^2(p).$$

D5

$$w_j = \frac{(\hat{\beta}_j - \beta_j)^2}{\sigma^2 q_{jj}} \sim \chi^2(1).$$

Confidence Intervals

- We use the previous distribution results to construct **exact** confidence intervals assuming that σ^2 is known. The more practical results, when σ^2 is unknown, are deferred to the next set of slides.
- Re write result D3 as

D3A

$$z_j \sim \frac{\hat{\beta}_j - \beta_j}{\sigma_{\hat{\beta}_j}} \sim N(0, 1)$$

where $\sigma_{\hat{\beta}_j}^2 = \sigma^2 q^{jj}$.

- Recall that when estimating the estimate of a parameter, it is a good idea to accompany it with some information about the the reliability of an estimator, or rather its unreliability—the extend to which it varies from sample to sample.
- The natural measure is the standard deviation of the estimator (or in more practical terms, its standard error.)
- This information can also be represented by a confidence interval.

- Using D3A, we have that

1 $\hat{\beta}_j \pm 1.96\sigma_{\hat{\beta}_j}$ is a 95% confidence interval for the unknown parameter β_j : You would read “the parameter β_j lies in the interval $\hat{\beta}_j \pm 1.96\sigma_{\hat{\beta}_j}$ with probability 95%”

2 $\hat{\beta}_j \pm 1.645\sigma_{\hat{\beta}_j}$ is a 90% confidence interval for the unknown parameter β_j .

3 $\hat{\beta}_j \pm 2.576\sigma_{\hat{\beta}_j}$ is a 99% confidence interval for the unknown parameter β_j .

- The higher the confidence level requested, the wider the interval.¹

- The cutoff values:

1 1.645 corresponds to the 90% confidence level.

2 1.96 corresponds to the 95% confidence level.

3 2.576 corresponds to the 99% confidence level.

¹The numbers 1.96, 1.645, and 2.576 do not appear by magic, they are cutoff values from the normal distribution, that is, if Z is a standard normal random variable:

$$\Pr(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1 - \alpha$$

where $1 - \alpha$ is the confidence level (example, if the confidence level is $1 - \alpha = 95\%$ then α is 5%)., and $z_{\alpha/2}$ is the cutoff value: the random variable Z is between between the values $-z_{\alpha/2} \leq Z \leq z_{\alpha/2}$ with probability $1 - \alpha$. Values for the normal distribution that are outside this interval happen with probability α .

Next steps: hypothesis testing

- In the next set of slides, we will construct hypothesis tests with the theory developed in these slides.
- We will still assume that σ^2 is known.
- Thus, the theory in this and the next set of slides is not implementable in practice because we usually don't know σ^2 , which needs to be estimated from the data.
- However, for pedagogic reasons, it is easier to start building hypothesis tests under the assumption of knowledge of σ^2 .
- The practical implementation of the tests, using the estimator $\hat{\sigma}^2$ is easily handled as an extension once the basic theory is developed.